

- Scientists investigated the production of sugars from crop waste for biofuel production. The scientists discovered that a strain of the fungus *Penicillium citrinum*, isolated from soil, was a good source of three different extracellular enzymes, **M**, **N** and **O**. These enzymes break down polysaccharides in cell walls.

The results are shown in Table 4.1.

enzyme	maximum activity /arbitrary units
M	292.83
N	111.72
O	6.54

The scientists carried out further research so that a solution containing the three enzymes (enzyme mixture) could be used for the most efficient production of sugars from crop waste.

[4]

- (b) Students investigated the composition of the cell wall of leaf cells of thale cress, *Arabidopsis thaliana*. The students began by isolating the cell wall components from the rest of the cell material.

The students used enzymes extracted from a fungal pathogen of *A. thaliana* to hydrolyse the cell wall components to smaller molecules.

The students prepared a reaction mixture containing the cell wall components and the enzymes.

After 24 hours, they separated and identified the smaller molecules found in the reaction mixture.

Four types of molecule were identified:

- short chains of β -glucose
- β -glucose
- peptides
- amino acids.

Explain the presence of these molecules in the reaction mixture after 24 hours of hydrolysis.

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[Total: 8]